Regression using Panel data for beginners

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Disclaimer: This article aims to describe panel data regression in the most simplest terms and is heavily based on the following books: Statistics II for Dummies, Econometrics For Dummies, and Introduction to Econometrics with R.

We all know what the very basic formula for regression looks like. It goes something like:

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$$Y = \beta_0 + \beta_1 X + \epsilon \qquad (1)$$

And we all know when to use regression as well. We use regression when we want to predict the value of something using certain factors, i.e., we want to find a linear relationship between two variables. If we use only one variable for predicting the value of the variable of our interest, like in eq(1), we call it simple linear regression. And if we use more than one variable, we call it as multiple linear regression. So before diving into regression for panel data directly, let's refresh our minds with the basics of simple and multiple linear regression.

Before we dive into regression itself, let's take a moment to think about if there is any other way to find a linear relationship between two variables, other than regression.

One way is to plot a scatterplot and find the answer visually. However, the caveat is that is is very easy to manipulate the scale of the axes and exaggerate a relation. We can also use a correlation coefficient such as Pearson's correlation coefficient. However, if we were to use only this to find the relationship between two variables, we might be able to get the quantity/degree of the correlation present, but not know for sure if it is linear in nature. Hence, a good way would be to use both the scatterplot as well as a correlation coefficient in conjunction.

Simple Linear Regression

A more robust and widely used method is simple linear regression as shown in eq(1), where Y is the dependent variable, X is the independent variable, beta_{0} is the y-intercept, beta_{1} is the slope(coefficient of the dependent variable), and epsilon is the error. This equation always provides you with an estimate and hence is normally written with a hat operator, but for the sake of this article, we will not be using the hat operator and it is understood that it is an estimate.

Once we know all that is required in writing the equation for the relationship between two variables of interest, we need to know how to find these coefficients and intercepts. The most common method used for this is the OLS method. Once we know how to write the equation and how to find the coefficients, we might think that we have completed our task in finding the best model. However, we need to make sure that two of the most important criteria/conditions are met before concluding that.

1. Once you have found the values of the coefficients as well the intercepts of your equation, it is important to check that the Y values you predict using your regression model have a normal distribution for each value of X. Something like this:

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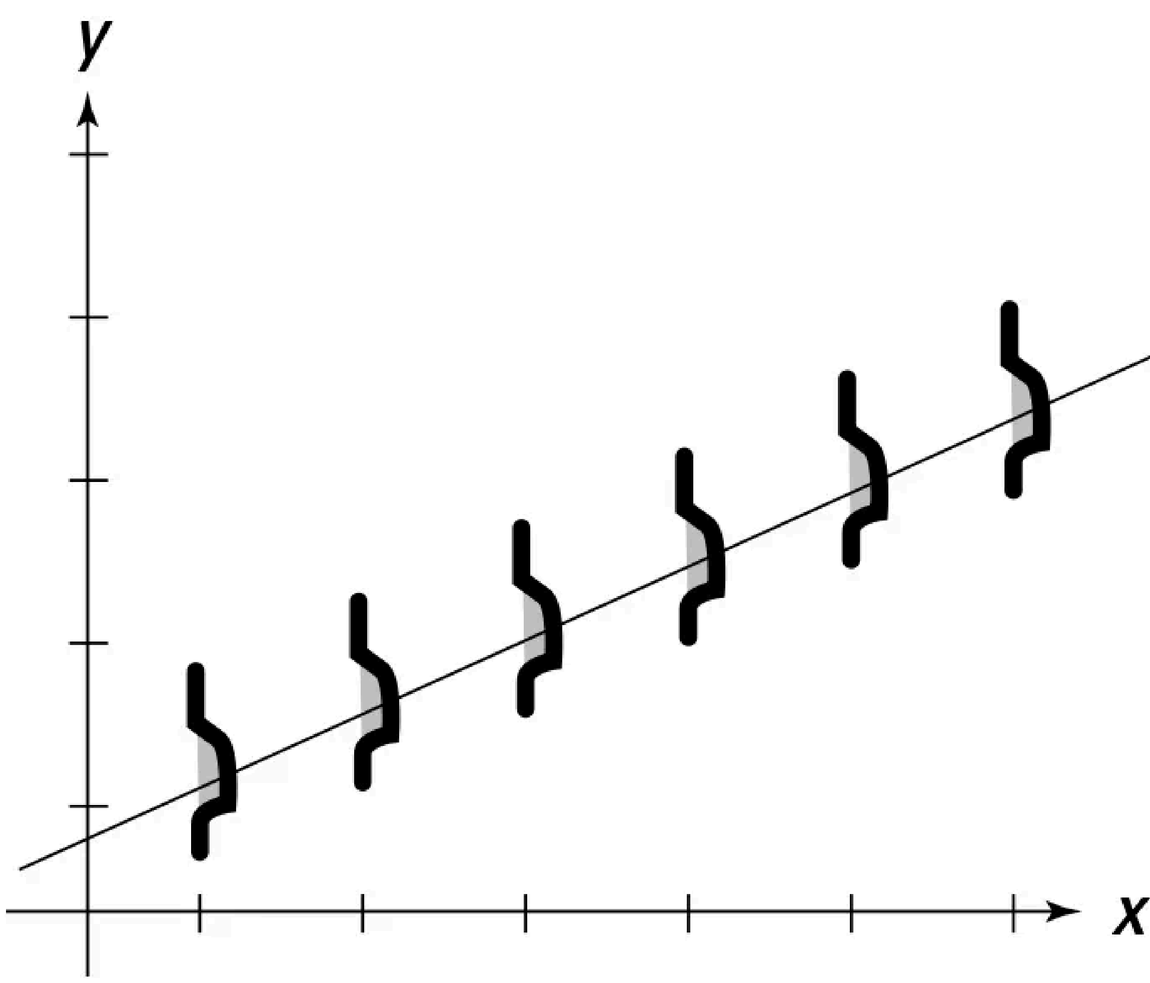
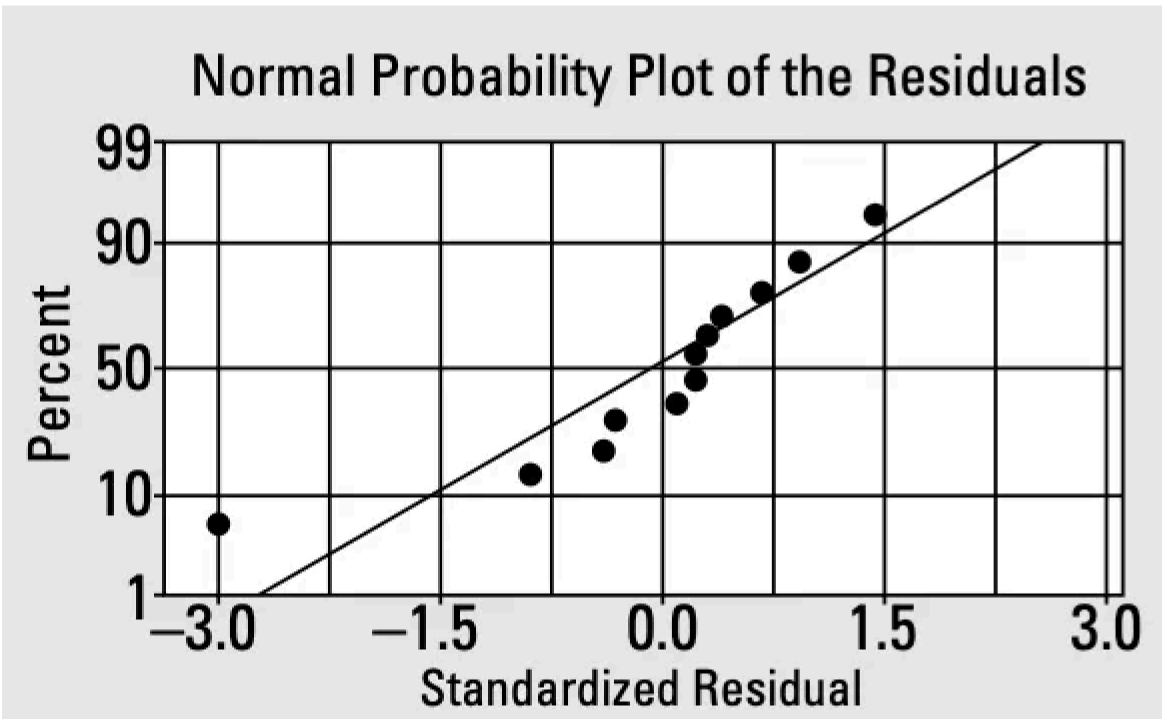


Credits: This figure is taken from the book Statistics II by Deborah Rumsey.

The predicted Y that is on the line is actually the mean of the population of the possible Y's for any X. So when we are inspecting the graph, we should see few of the data points falling above the line, few falling below the line, and mostly on the line. One way to visualize and confirm the normal distribution the Y values is to plot standardized residuals(errors) to see if most of them are close to 0. As can be seen in the following graph that the residuals are closer to 0, we can say that Y values have a normal distribution. One more thing to keep in mind is that the graph plotting the residuals should look random with no observable pattern at all.



Credits: This figure is taken from the book Statistics II by Deborah Rumsey.

2. The next thing to check is to see what happens to the y values when you move from left to right on the x-axis. The spread of the y values should be uniform throughout. This is known as the homoscedasticity condition. This is to ensure that when you find the best-fitting line, it works well for all values of x.

When you plot the standardized residuals against the fitted values, you can see that most residuals for all fitted values of y are closer to the line, except the very last, which in this case can be classified as an outlier.

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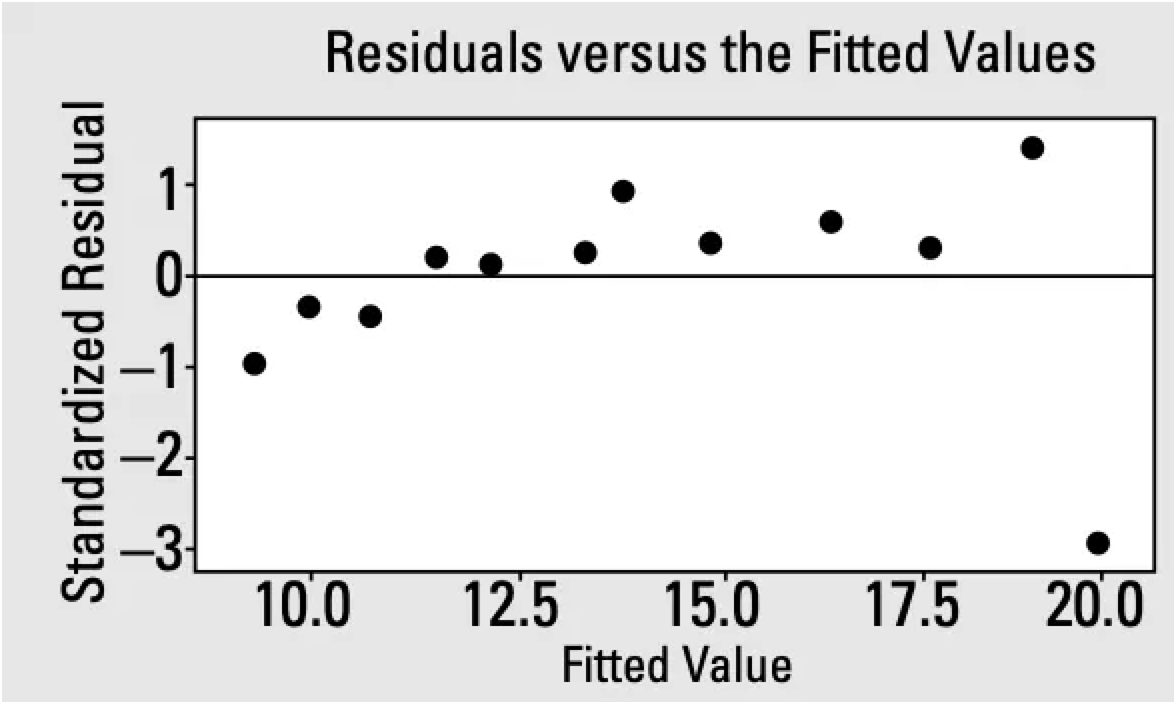

Credits: This figure is taken from the book Statistics II by Deborah Rumsey.

How do we measure the model fit?

Once we have established the equation, the algorithm to calculate the coefficients and the intercepts, and the necessary conditions that need to be satisfied for simple linear regression, it's time to see how we can quantify the fit of the line/model to our problem.

$$r^2$$

We measure the fit using the *coefficient of determination*, r^2. This number is measured in percentage, and is defined as the percentage of variability in Y present due to X. In simpler terms, the Y values that have been collected in the dataset, themselves have variability present in them. Once we choose a variable X to find it's relationship with Y, we are essentially using another variable X to help us explain that very inherent variability present in Y. So once the X values are plugged into the regression equation, we wanna find out how well it explains the already inherent variation in Y. Larger values of r^2 are good because that means X and its relationship with Y explain the differences in Y values pretty well, i.e., the line fits very well.

Multiple linear regression model

We have just refreshed our minds with what simple linear regression is. Now let's quickly look at multiple linear regression, and how it is different from simple regression. We all know, multiple linear regression looks like the eq(2), where we use more than one independent variable to explain the dependent variable.

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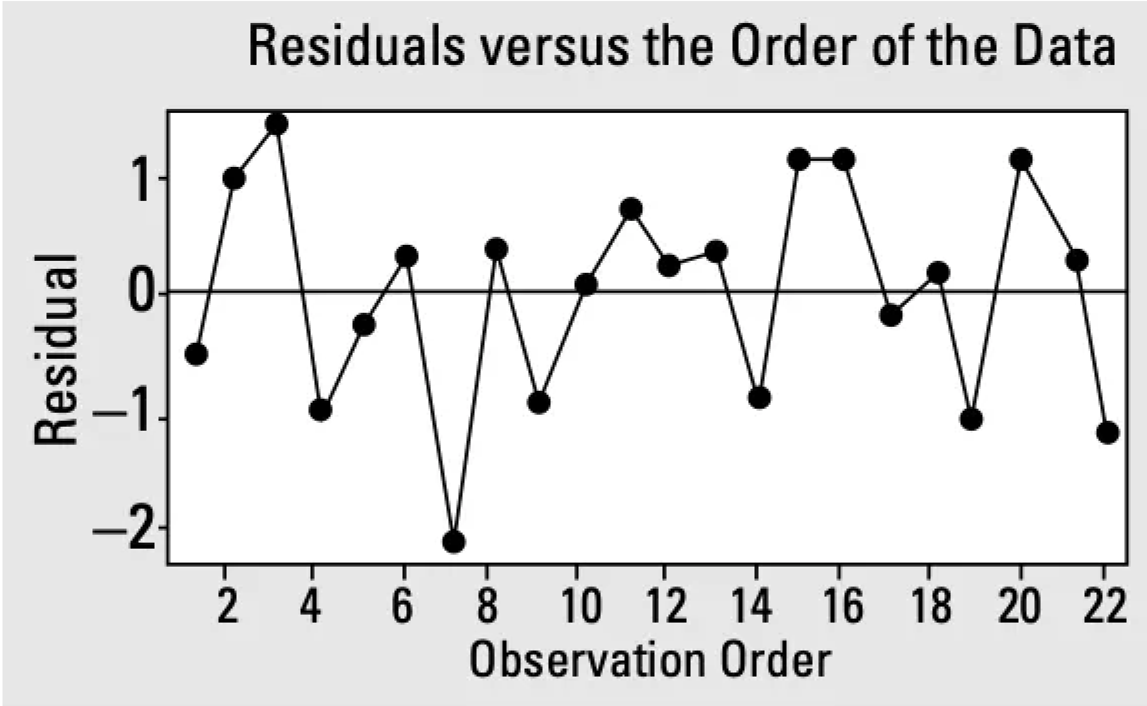
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ... + \beta_k X_k + \epsilon \qquad (2)$$

An important thing to keep in mind about multiple linear regression is a phenomenon called *multicollinearity*. This basically occurs when two or more X variables are correlated with each other. Apart from the fact that it leads to redundancy, it also leads to wrong and biased coefficients. Since two or more X variables are related, the algorithm might not be able to estimate the influence of one particular X variable on Y alone.

Just like simple linear regression calculates the influence of one independent variable on the dependent variable, multiple linear regression estimates the influence of each of the independent variables on the dependent variable, while keeping the influence of others constant. In other words, it calculates the *marginal contribution* of each independent variable.

As we saw in simple linear regression model, we need some necessary conditions to be met before we can declare the fit of the model. In the case of multiple linear regression model, including the two conditions already stated in simple linear regression, there is an additional condition as well.

1. *The residuals(errors) should be independent of each other*: What this means is that if we were to plot residuals by observation number(i.e., the order in which the observations came), we should not see any pattern in them. For example, a good graph between residuals and observation number would like this:



Credits: This figure is taken from the book Statistics II by Deborah Rumsey.

This can be a problem especially when you are working with time-series data, in which by it's very own nature, the previous data affects the next data and you might end up seeing a pattern.

Now that we have covered all the basics of simple linear regression and multiple linear regression, we can move to regression using panel data.

Panel Data

Panel data is defined the combination of cross-sectional data as well time-series data. In simple terms, for the dependent variable, we have a set of independent variables that affect it, that are collected over a period of time. So if we were to have *i* such units(observations), we would have *t* different values for each, one for each time period. An example of panel data that we will be using in this article is the [Fatalities dataset](#) from AER package.

This dataset reports the Traffic Fatalities for 48 states of the United States from the year 1982 to 1988. There are a total of 48 states, and we have observations for each 48 states over 7 years, making it a total 48 x 7 = 336 observations. There are a bunch of other factors in this dataset as well for each of the states over 7 years, but the one factor that we will be using in the following article is the Beer Tax implemented in each state in each of the 7 years, measured in dollars.

So why is regression different when it comes to panel data? That is because, it is very easy to find a relationship between two or three variables in a particular time period(year, month, day, etc.), but what happens when these relationships between the same two or three variables, exist over a period of time. How do we calculate the relationship then?

Regression for Panel Data

We are using the Fatalities dataset to understand panel data regression as described before. Let's start with the question: *How does the Beer Tax that is implemented in each state affect the number of Traffic Fatalities in that state, using the entire dataset?*

Remember that we have the same data for 7 different years. For starters, we can try out a simple regression model for two individual years separately. Let's say, we take 1982 and 1988. Using the R [code provided in the textbook](), we get the following simple regression models:

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$$FatalityRate = 2.01 + 0.15 BeerTax \qquad (1982) \qquad (3)$$
$$FatalityRate = 1.86 + 0.44 BeerTax \qquad (1988) \qquad (4)$$

Looking at the equations and the positive coefficients(slopes), we can see that these regression models indicate that there is a positive relation between Beer Tax implemented and the Traffic Fatalities, and that this effect increases in the year 1988, as compared to 1982. This is not something one would generally expect as the effect of Beer Tax implementation in a state.

This is a case of *omitted variable bias*, i.e., in this relationship we might have missed out on other important factors that could explain it better. One way to solve for this is to add more variables and go from simple regression to multiple regression. This will eliminate the omitted variable bias. But will that be enough?

An interesting aspect about panel data is the presence of *omitted unobservable factors*. What are these factors exactly? These factors can be something that is state-specific, meaning they differ for each state, but stay constant over time. For example, in our dataset, it can be something like population's attitude towards drinking. What can we do about these factors?

Fixed-effects regression:

This is where the fixed-effects regression comes into the picture. Let's look at the panel regression equation:

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$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + u_{it} \qquad (5)$$

Here*, i* represents the entity(state), and i = 1,2,…,n. Z_{i} represents the omitted unobservable variable also called as the *unobserved time-invariant heterogeneity*, meaning a factor that is specific to the individual entities, but does not change over time, and so is present for each i. Why will adding Z_{i} help us? Well, because in multiple regression we calculate the marginal contribution of each dependent variable, and hence can now estimate the correct value of \beta_{1}(i.e., effect of Beer Taxes on Y_{iy}) keeping the effect of the entity-specific unobservable variables(Z_{i}) constant.

Let’s simplify eq(5) using eq(6):

$$\alpha_i = \beta_0 + \beta_2 Z_i \qquad (6)$$

Equation 5 then becomes:

$$Y_{it} = \alpha_i + \beta_1 X_{it} + u_{it} \qquad (7)$$

What’s the significance of alpha_{i}? Essentially, alpha_{i}, i=1,2,…,n, is the set of individual intercepts, one for each entity(state), and can be thought of as the fixed-effect of entity i. We can also write the eq(7) as eq(8), if we were to write the same regression model with n-1 dummy variables. The coefficients of the dummy variables shift the regression function up or down, essentially giving it a new intercept for each entity i, as shown in the folllowing figure.

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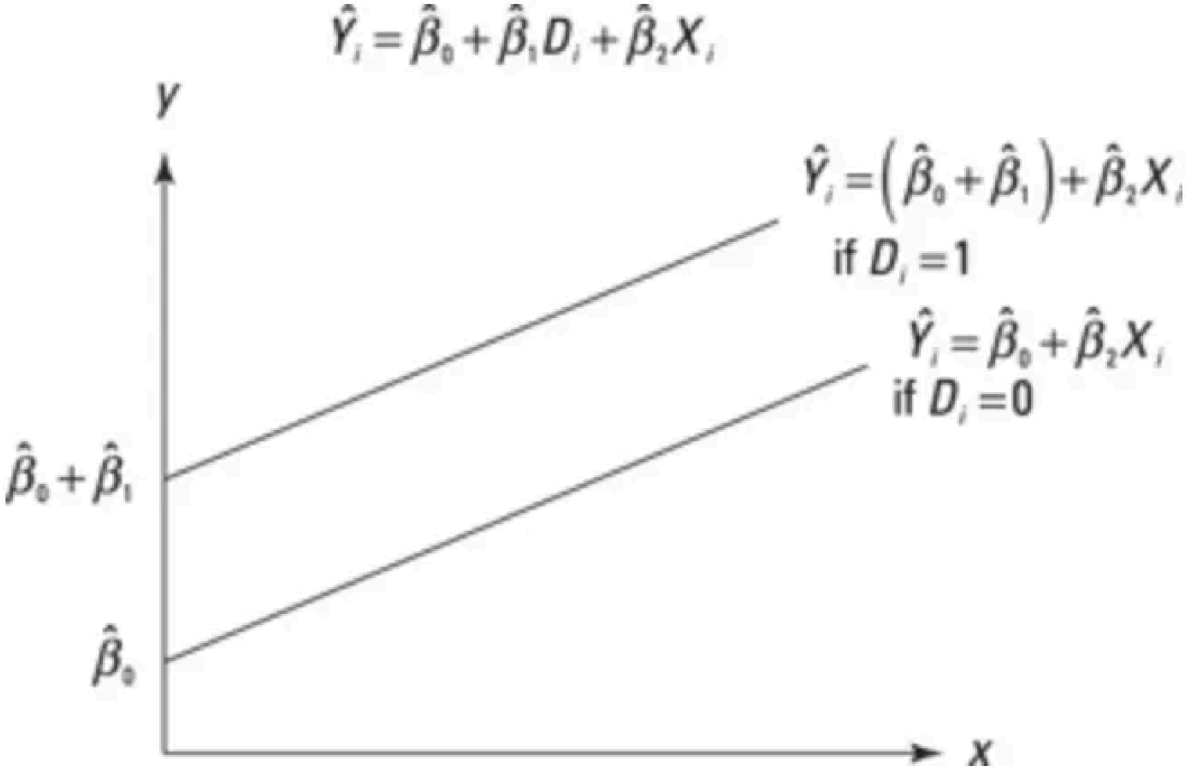

Credits: This figure is taken from the book Econometrics for Dummies.

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$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i \gamma_3 D3_i + ... + \gamma_n Dn_i + u_{it} \tag{8}$$

What if we have multiple X variables? In that case eq(8) simply becomes :

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Eq(9) is the equivalent of eq(7) for multiple variables, and eq(10) is the equivalent of eq(8), for *i=1,2,…,n* and *t=1,2,…,T.*

Given a standard simple or multiple regression model, we use the OLS algorithm to find the coefficients. But what about the situation where we have entity-specific/time-invariant unobservable variables as well? There are two ways we can go about it:

1. Using OLS as it is with the dummy variables(eq(8)).

2. Or using “entity demeaned” OLS: This is a computationally efficient method that is used by almost all the softwares. It is very simple and it follows like this:You start with taking averages on both sides of the eq(7), and then subtracting eq(12) from eq(7):

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$$\frac{1}{T}\sum_{t=1}^{T} Y_{it} = \beta_1 \frac{1}{T}\sum_{t=1}^{T} X_{it} + \alpha_i + \frac{1}{T}\sum_{t=1}^{T} u_{it} \qquad (11)$$

$$\bar{Y_{it}} = \beta_1 \bar{X_{it}} + \alpha_i + \bar{u_{it}} \qquad (12)$$

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$$Y_{it} - \bar{Y_{it}} = \beta_1 (X_{it} - \bar{X_{it}}) + (u_{it} - \bar{u_{it}}) \qquad (13)$$

$$\tilde{Y_{it}} = \beta_1 \tilde{X_{it}} + \tilde{u_{it}} \qquad (14)$$

With eq(14), we end up running simple regression on this without having to calculate n-1 other coefficients as well as the intercept, as we would have had to in the case of eq(8), hence, making it computationally faster. In the end we are getting the value of \beta_{1}, which is the point of interest here.

In our case of Fatalities Rate, after having considered entity-specific unobservable variables(time-invariant unobservable heterogeneity), our equation would become like these:

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$$FatalityRate_{it} = \beta_1 BeerTax_{it} + StateFixedEffects + u_{it} \qquad (15)$$

$$Fatali\tilde{t}yRate_{it} = \beta_1 Beer\tilde{T}ax_{it} + \tilde{u_{it}} \qquad (15)$$

Eq(15) is the fixed effects model with different intercepts for each entity and eq(16) is the “entity-demeaned ” version of the same. Let’s see what coefficients we get after applying OLS to both these equations. Using the R code provided in the book for this section, we see that we get \beta_{1} as -0.6659 in both cases. Hence, compared to eq(3) and eq(4), the coefficient for Beer Tax is negative and significant.

$$\beta_1 = -0.6659$$

Now we have arrived at the conclusion that for every dollar increased in the Beer Tax, there is a reduction of 0.66 per 1000 people. This may look a little too high of a ratio, so now the question arises, have we controlled for all the unobservable variables that might be affecting this coefficient.

Enter time-variant unobservable factors, meaning factors that don't vary across states but vary over time.

Time Fixed-Effects regression:

Let's continue with our eq(5), except let's assume we do not have unobservable variables that vary across entities, rather we just have unobservable variables that vary over time, but do not vary across states. This changes the eq(5) to:

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$$Y_{it} = \beta_0 + \beta_1 X_{it} + \delta_2 B2_t + ... + \delta_t BT_t + u_{it} \qquad (16)$$

Here, *B2, B3, …, BT,* are T-1 dummy variables, each corresponding to one of the T years(7 in our case). Again, *t = 1, 2, …, T*, and *i = 1, 2, …, n*. So in this way, we can get the right estimate of \beta_{1} while the effect of the dummy variables(time-variant unobservable variables) are held constant.

Now that we have the formula for controlling time-variant unobservable variables as well time-invariant(entity-specific) unobservable variables, let's see the regression equation that controls for both.

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$$Y_{it} = \beta_0 + \beta_1 X_{it} + \gamma_2 D2_i + \gamma_3 D3_i + ... + \gamma_i Dn_i + \delta_2 B2_t + ... + \delta_T BT_t + u_{it}$$

So our equation (15) becomes:

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$$FatalityRate_{it} = \beta_1 BeerTax_{it} + StateFixedEffects + TimeFixedEffects + u_{it} \qquad (17)$$

Now we can find the estimate of \beta_{1} while keeping both time-invariant as well as time-variant unobservable variables fixed. Using the R code provided in the book for this section, we see that \beta_{1} is calculated as -0.64, making our equation as:

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$$FatalityRate = -0.64 BeerTax + StateEffects + TimeFixedEffects \qquad (18)$$

Compared to eq(15), we see that \beta_{1} is pretty close to it. After having controlled for unobservable variables both time-variant and time-invariant in nature, we can confidently say that the value of the coefficient as -0.64 is not affected by *omitted unobservable variable bias.*

So to sum up, given a panel dataset and a question that is modeled as a regression problem, if we arrive at results for the coefficients that are contrary to common sense or theoretical expectations, we need to check for two things. *Omitted variable bias* and *omitted unobservable variable bias*. For the former, we need to add more independent variables and go for multiple regression. For the latter, we need to go for fixed effects model that controls for both time-variant and time-invariant unobservable variables.

References:

Statistics II for Dummies, Deborah Rumsey

Econometrics For Dummies, Roberto Pedace

Introduction to Econometrics with R, Christoph Hanck, Martin Arnold, Alexander Gerber, and Martin Schmelzer